Exploring the Genetic Underpinnings of Cognition in Schizophrenia Case-only Samples: A Meta-analytic Approach

Principal Investigator: Prof. Derek W. Morris, University of Galway, Ireland

Lead Analyst: Evie Doherty, University of Galway, Ireland

PGC Schizophrenia Cognition: ‘*g*’ Factor Protocol

The following protocol details the steps involved in deriving a general cognitive factor i.e. ‘*g*’, a statistically derived measure of within-person performance on a series of cognitive measures in your dataset.

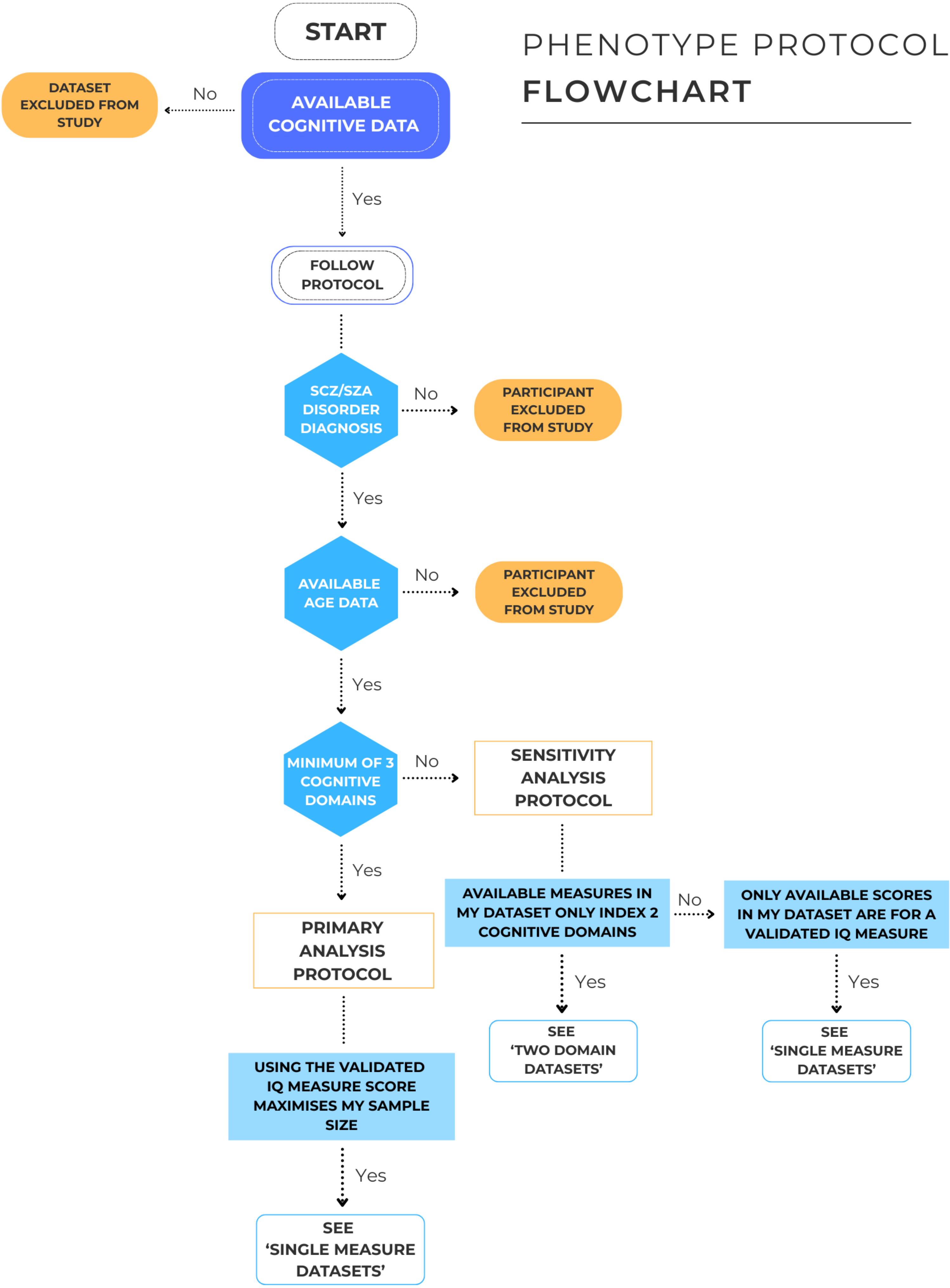
This protocol provides guidance on data preparation, outlier handling, age adjustment, score standardisation, and factor extraction. Guidance is also provided for datasets that do not meet the required criteria.

This protocol is intended for researchers analysing datasets which cannot be shared with is directly but still wish to contribute to the project. It is designed to promote consistency across participating cohorts. Any questions should be directed to e.doherty37@universityofgalway.ie.

A GitHub repository of code pertaining to this protocol has been created and can be accessed through the following link: https://github.com/eviedoherty/PGC_SCZ_Cognition_Protocol

The aim is to derive a ‘*g*’ factor by balancing two objectives: including the greatest number of available cognitive measures across as many cognitive domains as possible, while also maximising the number of participants with complete data.

PHENOTYPE PROTOCOL FLOWCHART



Step_01_Filter_Case_Only

The '*g*' factor should be restricted to participants with a consensus lifetime diagnosis of schizophrenia or schizoaffective disorder (SCZ/SZA) diagnosis according to the ICD (World Health Organization, 1992; WHO, 2019) and/or DSM criteria (American Psychiatric Association, 2000; APA, 2013).

Step_02_Filter_Relevant_Variables

The working data frame should be filtered to contain only the most relevant variables i.e. PGC ID, local ID, age, sex, cognitive measures etc. This step is not a requirement; however, it will allow you to work with a focused set of variables.

Step_03_Filter_Age

The '*g*' factor should be restricted to participants who have available age data. Age is required for the age-residualisation of cognitive measures (See Step_14_Age_Residualisation) and will be included as a control variable in the final analysis.

Step_04_Sex_Data

Where available, the distribution of sex in the sample should be assessed. However, this is not a requirement as sex data can be obtained from genotype files by the lead analyst during genetic analysis.

Step_05_Variable_Missingness

Calculate the percentage of missingness in each available cognitive measure, including the full-scale IQ measure (e.g. WAIS IQ, Ammons IQ, Shipley IQ) if present. This will allow you to evaluate data completeness and help determine which variables will contribute to your complete cases sample.

Please read the following carefully and refer to the flowchart.

Primary Analysis Datasets

For a dataset to be included in the main analysis, *at least one* measure from a minimum of three cognitive domains is required (e.g. Verbal Learning Task, list A for Verbal Learning & Memory; Continuous Performance Task for Attention; Trail-making A for Processing Speed). The 3-domain criterion was chosen for two reasons; 1) including measures that span three cognitive domains reduces the influence of domain specific characteristics which may occur when fewer domains are represented (Jensen, 1998), and 2) it is consistent with the methodologies used in prior comparable biobank scale studies (Trampush et al., 2017; Davies et al., 2018).

Note: Social cognition should not be included in deriving '*g*'. Although correlated, social cognition and neurocognition, i.e. general cognitive ability, are considered two separate areas of vulnerability in psychosis (Sergi et al., 2007).

Any dataset failing to meet the criteria for primary analysis (e.g. available cognitive measures index two cognitive domains, only has total scores for a valid IQ measure), see '**Sensitivity Analysis Datasets**'.

Step_06_Complete_Cases

Where a cognitive test has multiple subtests, each subtest is considered a *single* measure i.e. Verbal Learning Task List A Total, Verbal Learning Task List B, Verbal Learning Task Long-term Delayed Free Recall are all considered individual Verbal Learning & Memory measures. In the case that *only* the total score for a specific test is available (e.g. Continuous Performance Task total score rather than subtest scores such as identical pairs score or degraded stimulus score), the total score should be used.

Code Input: Cognitive measures should be assigned to domains based on the primary cognitive demand of the measure. Domains should be assigned using available information from the original test manuals and/or the MATRICS classifications (Nuechterlein et al., 2008; Kern et al., 2008). Six main cognitive domains should be used in this study; Verbal Learning & Memory, Visual Learning & Memory, Processing Speed, Attention, Working Memory, and Executive Function. Please note that several measures can index the same cognitive domain in a dataset.

Measures indexing verbal comprehension i.e. WAIS vocabulary, were not included in deriving the complete cases sample. Individuals with schizophrenia typically demonstrate relatively preserved performance on vocabulary tests despite exhibiting below average performance across other cognitive domains (Bombassaro et al., 2023).

Raw scores for each measure should be preferentially used. Where raw scores are unavailable, use scaled scores if the scaling procedure can be identified (e.g., WAIS subtest). Where both a raw score and a corrected raw score are present, i.e. omissions and/or commissions are incorporated, use the corrected score.

Code Output: A data frame containing the sample size for all possible conditions will be generated using the inputted measures. A condition refers to a unique combination of domains and measures contributing to the sample.

Each row will provide the following information for each condition; the number of samples (‘n’ column) when the minimum number of domains used is X (‘min_domains’ column), the actual number of domains included (‘actual_domains’ column) i.e. the minimum domains is 3 but the *actual* number of domains contributing to the sample size is 4, the exact domains contributing to the sample size (‘domains_used’ column), the number of measures included (‘n_tests’ column) and the exact measures contributing to the sample size (‘tests’ column; names as input by you).

Recommended:

1. Arrange the ‘n’ column is in descending order.
2. Obtain the sample size when *at least one* measure from a minimum of three cognitive domains is present (3 ‘min_domains’).
3. Additional measures and/or domains should be retained if the inclusion of each additional measure/domain reduces the sample size by no more than 1%.

Note: Where multiple combinations of cognitive domains and/or measures produce the same maximum sample size (i.e., contain an equivalent number of domains and measures but differ in the specific domains/measures), all protocol steps should be performed for all combinations.

IQ Measure: Where a total score for a validated IQ measure (e.g. WAIS IQ, Ammons IQ, Shipley IQ) is present in a dataset, the number of individuals with a score should be counted at this point. In the case that the number of individuals with a total IQ score exceeds the maximized sample size using the available cognitive measures (e.g., applying the three-domain criterion resulted in a sample of 179 individuals, while the use of the cognitive ability/IQ measure increased the sample to 289 individuals)., see ‘**Single Measure Datasets’.**

Step_07_Create_Complete_Cases_Dataframe

Create a new data frame containing complete cases using the cognitive measures identified in **Step_06_Complete_Cases**.

Step_08_Windsorise_Impossible_Values

Scores are considered outliers if they are outside the possible range of scoring on a given measure. The original test manuals should be consulted for the possible scoring ranges belonging to each cognitive measure. To maximise the number of complete case samples (Leys, et al, 2019), winsorising should be applied to impossible values. Winsorising requires replacing an outlying value in a measure with the nearest non-outlying value for that measure (Wilcox, 2005).

Step_09_Winsorise_Outliers

Scores were also classified as outliers if they fell more than ±3 standard deviations from the mean. The following code calculates the mean, standard deviation, and the upper and lower cutoff values (±3 SD) for each cognitive measure ($cutoffs). It also generates a list of participants with scores identified as outliers for winsorisation ($participant_results) and a table comparing the original and winsorised values for all modified scores ($replacements).

Step_10_Normality_Skewness

Normality and skewness should be assessed using histograms and QQ-plots.

Binary measures or those demonstrating total floor or ceiling effects should be removed as they provide limited data on the variability in cognitive performance (Gill et al., 2008). Where measures are resultantly removed from a dataset, **Step_06_Complete_Cases** through **Step_10_Normality_Skewness** should be re-run without those measures.

Step_11_Transformations

Appropriate transformations should be applied (i.e. log, square-root) where measures demonstrate a moderate-severe positive or negative skew. Determining the appropriate transformation requires considering the type of data you are working with (e.g. count or continuous) and the severity of the skew.

Reflection should be applied to negatively skewed data prior to transformation.

Step_12_Orient_Scores

All scores should be oriented so that higher scoring indicates better cognitive ability.

Step_13_Age_Residualisation

To adjust for age-related cognitive differences, the raw scores should be age-residualised using a multiple regression analysis. Where a scaled cognitive measure score is already age-adjusted, age-residualisation is not required.

Step_14_Z_Scoring

The age-adjusted cognitive measure scores in each dataset should be standardised by generating z-scores.

Step_15_Cronbach_Alpha

A Cronbach's Alpha coefficient should be calculated to assess the internal reliability of the cognitive measures within each dataset. The internal consistency of a dataset should be >0.70 (std.alpha). Individual cognitive measures should be dropped from a dataset where the corrected item-total correlation is <0.30 (r.drop).

Step_16_Run_PCA

A Principal Component Analysis (PCA) should be used to generate '*g*' using the z-scored cognitive measure scores for the complete cases in each dataset. The first unrotated component should be extracted from the PCA and used as an estimate of '*g*' i.e., general cognitive ability.

To ensure '*g*' is interpretable and representative of a single dominant component, the first principal component should be further examined. The proportion of variance explained by the first unrotated principal component should be >40%, as per prior literature (Deary et al., 2010). All cognitive measures should sufficiently load on to the first unrotated principal component (>0.40). A parallel analysis should be used to statistically confirm whether the first unrotated principal component suitably estimates '*g*'. The parallel analysis will provide a scree plot and determine the optimal number of factors that should be retained for your dataset. Where the parallel analysis suggests a multi-component solution, the first principal component should still be extracted as an estimate of '*g*', if the proportion of variance explained is >40%.

Note: For the datasets with several combinations of measures/domains producing the same maximum sample, the combination with the first principal component accounting for the greatest proportion of variance should be selected and used in the final analysis.

Sensitivity Analysis Datasets

Any dataset failing to meet the criteria for primary analysis (i.e. three-domain rule) will be included in a subsequent sensitivity analysis and the results compared to the primary analysis.

Two Domain Datasets

Where the available cognitive measures in a dataset only index two cognitive domains, a ‘*g*’ factor should still be derived using the ‘*g*’ factor protocol for ‘Primary Analysis Datasets’.

Single Measure Datasets

Where the only available scores in a dataset are for a validated IQ measure (e.g. WAIS IQ, Ammons IQ, Shipley IQ), the following steps of the protocol should be applied.

- **Step_01_Filter_Case_Only** through **Step_05_Variable_Missingness**
- **Step_08_Windsorise_Impossible_Values** through **Step_14_Z_Scoring**

In the case that the number of individuals with an IQ total score exceeds the maximized sample size using the available cognitive measures (as per **Step_06_Complete_Cases**), the dataset should be treated as a ‘Single Measure Dataset’.

Note: Age-residualisation (Step_14_Age_Residualisation) may not be required as most cognitive ability/IQ measure scores are age-scaled. Consult the original test manual.

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